

Harsh Shroff

667-464-5255 | harshshroff@gmail.com | <https://www.linkedin.com/in/harshshroff/> | <https://github.com/HarshShroff/>

SUMMARY

Innovative **AI/ML Engineer, Data Scientist, and Data Analyst** with expertise in **large-scale data processing, machine learning, generative AI, deep learning, and data engineering** for solving complex real-world challenges. Proven experience in **building and optimizing AI-powered systems, automating workflows, and deploying ML models at scale**. Adept at **data modeling, visualization, and statistical analysis** with a strong command of **cloud platforms, databases, and MLOps frameworks**. Passionate about contributing to **advanced ML research** in **defense and healthcare** domains by developing **scalable AI-driven solutions**.

SKILLS

Machine Learning & AI: Generative AI (*LLMs, RAG, LangChain, LlamaIndex*), Deep Learning (*TensorFlow, PyTorch, YOLOv8, OpenCV*), Natural Language Processing (NLP), Computer Vision (CV), Time Series Forecasting, Predictive Modeling, MLOps (*MLflow, Kubeflow, Vertex AI*)

Data Engineering & Cloud Computing: AWS (*Lambda, EC2, S3, Textract, Comprehend, Glue, Redshift, Kinesis, Bedrock*), SQL & NoSQL Databases (*PostgreSQL, MySQL, MongoDB, Elasticsearch*), Data Pipelines (*ETL, Airflow, Terraform, Spark, Hadoop*)

Data Analysis & Visualization: Python (*Pandas, NumPy, SciPy, Seaborn, Matplotlib*), BI Tools (*Tableau, Power BI, Streamlit, Dash*), Statistical Analysis (*A/B Testing, Regression, Hypothesis Testing*)

Software Development & Automation: Python, R, SQL, C++, Embedded C, API Development (*FastAPI, Flask*), DevOps (*CI/CD, Docker, Kubernetes, Terraform*), Agile Methodologies, Git, Version Control

EXPERIENCE

Data Analyst | ML Engineer

[VITG Corp.](#), Halethorpe, MD | Jun 2024 - Present

- Developed a **resume scoring Slack bot** powered by **Llama LLM** and **AWS services (Lambda, EC2, S3)** to automate candidate screening, reducing manual processing by **60%**.
 - Built a **chatbot-driven geospatial tool** to match addresses with government conservation zones, aiding **environmental planning** and **property regulations in Hyderabad**.
 - Automated geospatial data extraction** from **2500+ scanned PDFs/images** using **OCR-based lat-long extraction** and **Claude 3.5 Haiku LLM**, visualizing results on interactive maps.
 - Designed and **deployed an AI-powered web-app** using **Streamlit** and **AWS** for real-time geospatial analysis, improving **data accessibility by 30%**.
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Data Researcher – AI & Robotics

[Center for Real-time Distributed Sensing and Autonomy](#), UMBC | Mar 2023 – Dec 2024

- Led a team of 5 to develop **AI-powered multi-modal robotic navigation** for *Boston Dynamics Spot, Clearpath Husky & Jackal robots* under the [ArtIAMAS project](#) (funded by the **US Army Research Lab & DEVCOM**).
 - Implemented **gesture-based robotic controls** using **Microsoft HoloLens 2** and **Python**, enabling **intuitive human-robot interaction** for the “*Human Machine Teaming*” project.
 - Optimized **LIDAR-based pathfinding algorithms** using **reinforcement learning**, reducing navigation errors by **20%** in dynamic environments.
 - Designed a **GPS-based navigation system** for **Clearpath Husky robots**, achieving **95% path accuracy**.
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Data Science Intern

[Freshwater Institute – The Conservation Fund](#), Shepherdstown, WV | Aug 2023 - Dec 2023

- Achieved a 92% accuracy rate in fish fillet color grading using **Python, OpenCV & YOLOv8 Classifier model** deployed on Raspberry Pi 4.
 - Designed and developed an in-house 3D printed portable case for a **real-time assessment** on fillet grading in aquaculture.
 - **Co-authored a research paper** titled “[Fish Fillet Color Profiling Using Deep Learning](#)”, detailing the application of AI in **precision aquaculture and food safety**.
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EDUCATION

Master’s in Data Science

[University of Maryland Baltimore County \(UMBC\)](#) | Aug 2022 - May 2024

GPA: 3.8/4.0

Relevant Coursework: Natural Language Processing, Data Analysis & Visualization, Data Management, Big Data Processing, Financial Data Science, Machine Learning, Artificial Intelligence

PROJECTS

Document Parsing for Environmental Planning

Tech Stack: AWS (Textract, S3, Redshift), Python, Tableau

- **Automated geospatial data extraction** from 2500+ scanned PDFs using **OCR and ML models**, increasing data processing efficiency by **40%**.
- Developed interactive **Streamlit dashboard**, providing stakeholders with **real-time insights**, and enabling **faster decision-making** on conservation projects.

Dashboard for Autonomous Robotic Patrols

Tech Stack: Python, ROS, LiDAR, Deep Learning, Computer Vision

- Developed a **real-time asset monitoring dashboard** using **Python, Dash** and **ROS**, tracking *battery health, network connectivity, and asset movement via GPS* in **military simulations**.
- Integrated **real-time object detection** using **OpenCV** and **Python** and **obstacle detection** via **LiDAR**, **enhancing robotic autonomous movement and safety**.

AirbnbLense – Personalized Recommendation System, [GitHub](#)

Tech Stack: Python, OpenAI API, Computer Vision, Tableau, Big Data

- Built a **recommendation engine** combining **structured and unstructured data** (160MBs of CSV, 130GBs of images).
- Implemented **computer vision models** to analyze listing images for **style, color, and object recognition**.
- Reduced **Airbnb bounce rates** by 20% and **improved engagement** by 40% through tailored recommendations.

Cyclists’ Performance Dashboard, [GitHub](#)

Tech Stack: Python (Dash, Seaborn), Machine Learning

- Developed an **interactive dashboard** to track **real-time cyclist performance metrics** using **Python** and **Dash**, improving **training efficiency** by 20%.
 - Implemented **predictive analytics** for **personalized training insights**, enhancing **athlete performance**.
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ACHIEVEMENTS

- **AWS Machine Learning Foundations Certification** – Proficiency in core ML concepts & AWS tools.
- **Published research in IEEE Xplore** – “[Mosquito Species Identification using TinyML Devices](#)”.
- **1st place, ITU International Hackathon** – Developed a **TinyML device** on **Mosquito Species Identification**.
- Created a **LinkedIn Job Analyzer Chrome Extension** – Uses **Perplexity AI API** for job insights.